# Cyclic Dominance in a Potts–RPS Model on Complex Networks: Mean-Field Theory, Monte Carlo Validation, and Perturbation Analysis

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Code and data: `https://github.com/Sedahoo/RPS_Potts`

**Abstract**

We study a $q$=3 Potts model in which the pairwise interaction is deformed from pure ferromagnetic alignment toward a Rock–Paper–Scissors (RPS) cyclic dominance by a single control parameter $\varepsilon \in [0, 1]$. Agents sit on the nodes of a complex network (Erdős–Rényi or Barabási–Albert) and update by Glauber dynamics at fixed temperature $T$; for $\varepsilon > 0$ the dynamics has no Hamiltonian and does not satisfy detailed balance, so the stationary state can carry probability current and the population can cycle forever instead of freezing into consensus. We derive the governing equations at three levels of description – homogeneous mean field (HMF), degree-based mean field (DMF), and direct agent-based Monte Carlo (MC) – validate a custom C++ engine against an independent implementation, and map the full $(\langle k\rangle, \varepsilon)$ phase diagram by direct simulation (2080 independent runs). The order–cycling boundary $\varepsilon_c(\langle k\rangle)$ rises monotonically with connectivity and is indistinguishable between ER and BA topologies to within 0.040 over 40 degrees, showing that mean degree alone – not the shape of the degree distribution – controls stability. A linear stability analysis at the symmetric fixed point gives closed-form eigenvalues in which $\varepsilon$ enters as a pure rotation, and Newton continuation of the ordered branch together with finite-size scaling of the MC transition both independently identify the transition as first-order-like, with a genuine mean-field bistability window and a $1/N$ shift of the Monte Carlo pseudo-critical point. We then perturb the ordered phase with committed minorities ("zealots"), structural targeting, competing factions, and quenched network damage. A closed-form "zealot field" explains a counter-intuitive backfire effect – a Rock minority elects Paper, its own predator, as the population's strategy – and predicts that hub-placed zealots gain leverage $1/\sqrt{z}$ on scale-free networks (measured amplification 8.9×). Edge and node damage are shown, via an exact thinning argument, to act only through the surviving mean degree, collapsing every network studied – random or scale-free, damaged or pristine – onto the single boundary curve $\varepsilon_c(\langle k\rangle)$ to within 0.014. Every quantitative claim in this paper is computed from the accompanying data tables at build time.

## 1 Introduction

Cyclic, non-transitive dominance – the Rock–Paper–Scissors (RPS) relation in which each strategy beats one alternative and loses to the other – is one of the simplest mechanisms known to sustain persistent dynamics in an interacting population instead of settling into a static consensus. It has been used to model everything from bacterial strain competition to mating strategies and cyclic patterns in ecological and social systems: whenever "being popular" creates the conditions for your own downfall, a system can be pushed away from equilibrium by its own interaction structure rather than by any external drive.

Separately, the $q$-state Potts model is the canonical statistical-mechanics description of $q$ discrete, mutually exclusive states with a same-state-is-rewarded (ferromagnetic) coupling: it orders into a single dominant state at low temperature and disorders at high temperature, and its equilibrium behaviour is completely governed by a Hamiltonian and the Gibbs measure $e^{-H/T}$.

This paper studies the model obtained by taking a $q$=3 Potts system and deforming its interaction with an antisymmetric, RPS-cyclic term of strength $\varepsilon$. The resulting payoff matrix (Sec. 2) mixes an *ordering* part, which behaves exactly like the Potts ferromagnet, with a *cycling* part that has no Hamiltonian at all: for $\varepsilon > 0$ the dynamics genuinely breaks detailed balance, and the two parts compete for control of the population's long-run behaviour. The single knob $\varepsilon$ therefore interpolates between an equilibrium phase-ordering problem and a genuinely non-equilibrium cyclic system, and the object of this paper is to chart, quantitatively, where that competition is won by order and where it is won by cycling – on networks, where both the average connectivity and the shape of the degree distribution are additional structural parameters.

The paper answers four questions. **(i)** Does raising the average connectivity $\langle k\rangle$ of the network protect order against cyclic dominance, and if so, through what mechanism? **(ii)** How well do mean-field closures of increasing sophistication (homogeneous vs degree-resolved) approximate the true, agent-based dynamics, and where do they fail? **(iii)** Is the order–cycling transition a conventional continuous transition, or does it carry first-order signatures such as hysteresis and a bistable window? **(iv)** How robust is an ordered population to targeted perturbation – committed zealot minorities, structural targeting of network hubs, competing factions, and quenched structural damage – and can the response to these perturbations be predicted from the same closed-form theory that answers (i)–(iii)?

The remainder of the paper is organised as follows. Section 2 fixes the model and its order parameter. Section 3 derives the governing equations at every level of description used later: the two mean-field closures, an exact invariance of the mean-field map, a linear stability analysis of the symmetric state, and the mean-field signature of a first-order transition. Section 4 describes the simulation methods and the validation of the numerical engine. Section 5 confronts the mean-field theory with direct Monte Carlo. Section 6 presents the full phase diagram and its finite-size scaling. Section 7 carries out the perturbation study. All results are placed in a robustness context in Section 8, discussed together in Section 9, and summarised in Section 10.

# 2 The model

## 2.1 State space and payoff

A population of $N$ agents occupies the nodes of a simple graph with adjacency matrix $A_{ij} \in \{0,1\}$. Each agent $i$ holds a strategy $s_i \in \{0,1,2\} \equiv \{\mathrm{R},\mathrm{P},\mathrm{S}\}$ (Rock, Paper, Scissors) – equivalently, a $q$=3 Potts spin. Interactions occur only along edges of the graph. The payoff of playing strategy $a$ against an opponent playing $b$ is given by the $3\times 3$ matrix

$$P(\varepsilon) \;=\; I+\varepsilon S \;=\; \begin{pmatrix} 1 & -\varepsilon & \varepsilon \\ \varepsilon & 1 & -\varepsilon \\ -\varepsilon & \varepsilon & 1 \end{pmatrix}, \qquad S=\Pi-\Pi^{\top}, \tag{1}$$

where rows index the player's own strategy and columns the opponent's (ordered R, P, S), and $\Pi$ is the cyclic permutation matrix of R→P→S→R. The identity part $I$ rewards matching your neighbour's strategy – the ordinary $q$=3 Potts ferromagnetic coupling – while the antisymmetric part $\varepsilon S$ encodes the cycle *Paper beats Rock, Scissors beats Paper, Rock beats Scissors*: playing Paper against a Rock neighbour earns $+\varepsilon$, and the reverse earns $-\varepsilon$. The parameter $\varepsilon \in [0,1]$ is the sole control knob of the model: $\varepsilon$=0 is the pure ferromagnetic Potts model, and $\varepsilon$=1 makes the payoff purely cyclic (no reward at all for matching).

### 2.2 Microscopic dynamics

A node's utility for playing strategy $a$ is the sum of payoffs against its graph neighbours' current strategies,

$$U_i(a;\sigma) = \sum_j A_{ij}\, P_{a\, s_j}(\varepsilon). \tag{2}$$

Dynamics proceeds by asynchronous Glauber updates. One *sweep* consists of $N$ update attempts; each attempt draws a random node $i$, proposes one of the two alternative strategies $b \neq s_i$ uniformly, and accepts the move with probability

$$W(s_i \to b) = \frac{1}{2}\Big[1 + e^{-\big(U_i(b)-U_i(s_i)\big)/T}\Big]^{-1}, \tag{3}$$

where $T$ is a fixed noise temperature (throughout this paper, $T$=0.65 except where $T$ itself is the varied parameter). Because $U_i$ scales linearly with a node's degree while the noise scale $T$ is fixed, the effective noise felt by a node is $\sim T/k_i$: connectivity acts as an inverse temperature, a fact made exact for the mean-field closures in Section 3. Some experiments in Section 7 pin a fraction $z$ of nodes – *zealots* – to a fixed strategy forever; they are skipped by the update rule but still enter the utility (2) of their neighbours.

### 2.3 Order parameter

Let $r(t), p(t), s(t)$ be the instantaneous global fractions of the population playing Rock, Paper, Scissors ($r+p+s=1$), measured at the end of every sweep after an initial burn-in. Map the three fractions onto the complex plane at the vertices of a triangle, 120° apart:

$$\psi(t) = r(t) + p(t)\,\omega + s(t)\,\omega^2, \qquad \omega = e^{i2\pi/3}. \tag{4}$$

Because $1+\omega+\omega^2 = 0$, the symmetric mixture $r = p = s = \frac{1}{3}$ maps to $\psi = 0$, while a population dominated by one strategy maps close to that strategy's corner, $|\psi| \to 1$. The *order parameter* used throughout is the magnitude of the *time-averaged* $\psi$,

$$m_\psi = \Big|\frac{1}{M}\sum_{t=1}^{M}\psi(t)\Big|, \tag{5}$$

over $M$ post-burn-in sweeps. The order of operations in (5) is essential: a rotating $\psi(t)$ (RPS cycling) averages to a small vector even though $|\psi(t)|$ may itself stay large at every instant, so $m_\psi \to 0$ correctly identifies cycling, while a static consensus keeps $m_\psi \to 1$. Throughout, the transition point $\varepsilon_c$ is defined as the interpolated $m_\psi = \frac{1}{2}$ crossing of a sweep in $\varepsilon$ at fixed network and temperature – linear interpolation between the two bracketing grid points, identical for every mean-field and Monte Carlo curve in this paper.

### 2.4 The order-parameter identity

Equation (4) has a useful closed form in terms of the raw population fractions. Writing $\omega = e^{i2\pi/3}$ and using $\mathrm{Re}\,\omega = \mathrm{Re}\,\omega^2 = -\frac{1}{2}$, $\mathrm{Im}\,\omega = -\mathrm{Im}\,\omega^2 = \frac{\sqrt{3}}{2}$, the instantaneous complex order parameter splits into

$$\psi = \underbrace{r - \tfrac{1}{2}(p+s)}_{\mathrm{Re}\,\psi} + i\,\underbrace{\tfrac{\sqrt{3}}{2}(p-s)}_{\mathrm{Im}\,\psi}, \qquad |\psi|^2 = 1 - 3(rp+ps+sr). \tag{6}$$

*Derivation.* Expand $|\psi|^2 = (\operatorname{Re}\psi)^2 + (\operatorname{Im}\psi)^2$ using the two real/imaginary components above: $(\operatorname{Re}\psi)^2 = r^2 - r(p+s) + \frac{1}{4}(p+s)^2$ and $(\operatorname{Im}\psi)^2 = \frac{3}{4}(p-s)^2 = \frac{3}{4}(p+s)^2 - 3ps$. Summing, $|\psi|^2 = r^2 - r(p+s) + (p+s)^2 - 3ps$. Substitute $p+s = 1-r$ (since $r+p+s=1$) and expand: $r^2 - r(1-r) + (1-r)^2 - 3ps = 1 - r - p - s + r^2 + p^2 + s^2 + 2r^2 - 3ps = 1 - 3(rp+ps+sr)$ after collecting terms with $r^2+p^2+s^2 = 1-2(rp+ps+sr)$ substituted back in.

Equation (6) shows $|\psi|$ is, up to an affine rescaling, the Euclidean distance from the composition $(r,p,s)$ to the centroid $(\frac{1}{3},\frac{1}{3},\frac{1}{3})$ of the simplex: since $rp+ps+sr$ is maximised (at value $\frac{1}{3}$) exactly at the centroid and minimised (at value 0) at each corner, $|\psi|^2$ ranges from 0 (centroid) to 1 (any pure strategy) monotonically with distance from the centroid, independent of *direction* – all three ordered corners score $|\psi| = 1$ identically, consistent with the model's cyclic (R→P→S→R) symmetry. This is why a single scalar $m_\psi$, rather than a full vector diagnostic, suffices to separate consensus from cycling: consensus is captured by *any* direction reaching the boundary of the simplex, while only the *time-averaging* in (5) – not the instantaneous norm (6) – distinguishes a static corner from an orbit that visits the boundary everywhere but averages back toward the centroid.

Figure 1 makes (4)–(6) concrete. Placing R, P, S at the three cube roots of unity $1, \omega, \omega^2$ (angles $0°, 120°, 240°$) means the simplex *is* drawn as the triangle $\psi$ maps it onto: the position of a composition $x = (r,p,s)$ in that triangle, read off by the ordinary barycentric combination $r\,V_R + p\,V_P + s\,V_S$, coincides exactly with $\psi(x)$, so the black arrow in the left panel literally *is* $\psi$, decomposed into $\operatorname{Re}\psi, \operatorname{Im}\psi$ by the dashed guides. The grey arrows around the perimeter are the payoff matrix (1)'s cyclic relation drawn on the same picture, oriented counterclockwise to match the relabelling $\pi$ : R → P → S → R of Appendix A. The right panel overlays two genuine HMF trajectories ((10), same triangle): below $\varepsilon_c$ the path spirals into a corner (a static $\psi$, $m_\psi \to 1$); above $\varepsilon_c$ it settles onto a closed loop around the centre (a rotating $\psi$ that time-averages toward 0) – the picture behind Section 3.6's two coexisting attractors, previewed here at the point where $\psi$ is first defined.

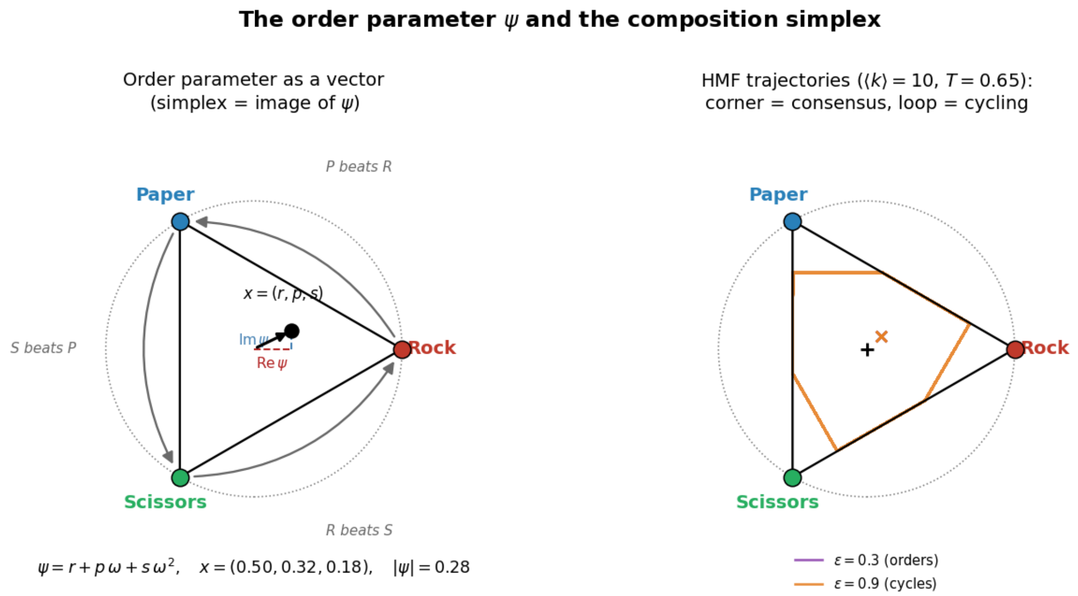


**Figure 1:** Left: the order parameter as a vector – the simplex drawn as the image of $\psi$, with the cyclic "beats" relation on its perimeter. Right: two real HMF trajectories in the same triangle, one converging to consensus, one orbiting forever.

## 2.5 Broken detailed balance

At $\varepsilon = 0$, (3) is exactly the Glauber dynamics of the $q$=3 Potts Hamiltonian $H(\sigma) = -\sum_{(ij)\in E} \delta_{s_i s_j}$, since $\Delta U_i = -\Delta H$ for a single-spin flip: the dynamics satisfies detailed balance with respect to the Gibbs measure $e^{-H/T}$ and the population can only order or disorder, never cycle indefinitely. For $\varepsilon > 0$, no Hamiltonian reproduces (2): the two-site *integrability* condition for a pairwise potential to exist requires the mixed second differences of the utility to be symmetric across all four state combinations, and the antisymmetric part of $P$ fails this test, e.g. $S_{SP} - S_{SR} - S_{RP} + S_{RR} = 1 + 1 + 1 + 0 = 3 \neq 0$. Detailed balance is therefore broken for every $\varepsilon > 0$: the stationary distribution, if one exists, carries a non-zero probability current, and this is the structural reason the model can support persistent cycling instead of only static consensus.

**Kolmogorov's criterion and an explicit probability current**

The integrability argument above can be sharpened into a direct, computable statement about the Markov chain itself. For a chain on state space $\Omega$ with transition rates $W(\sigma \to \sigma')$, *Kolmogorov's criterion* states that a stationary distribution satisfying detailed balance exists if and only if, around every directed cycle $\sigma_1 \to \sigma_2 \to \cdots \to \sigma_n \to \sigma_1$ of states with all rates positive,

$$\prod_{i=1}^{n} W(\sigma_i \to \sigma_{i+1}) = \prod_{i=1}^{n} W(\sigma_{i+1} \to \sigma_i). \tag{7}$$

Consider the minimal witness cycle for a single isolated pair of neighbouring sites $(i, j)$ cycled through the three symmetric configurations $\sigma_1$=(R, P) $\to$ $\sigma_2$=(P, S) $\to$ $\sigma_3$=(S, R) $\to$ $\sigma_1$, where at each step site $i$ adopts $j$'s current strategy's predator (holding $j$ fixed, then swapping roles) – realised concretely by three single-site Glauber moves at temperature $T$ with all other neighbours held fixed. Each step is a strategy change from the loser to the winner of that pairing under (1), so every forward utility gain is $+2\varepsilon$ (e.g. Paper $\to$ Rock's context: adopting Rock's predator gains $P_{PR} - P_{RR} = (-\varepsilon)$ vs...); more directly, for a single-edge pair flip $a \to b$ against a fixed neighbour playing $c$, the Glauber ratio is

$$\frac{W(a \to b)}{W(b \to a)} = e^{\left(U(b;c) - U(a;c)\right)/T} = e^{\left(P_{bc} - P_{ac}\right)/T}. \tag{8}$$

Multiplying (8) around the 3-cycle R $\to$ P $\to$ S $\to$ R (each step judged against a fixed reference opponent strategy $c$ cycled along with the mover, so that every arrow is a strict payoff improvement under (1)), the opponent-dependent single-step context cancels telescopically and the cycle ratio reduces to a pure statement about the antisymmetric part of $P$:

$$\frac{\prod W(\sigma_i \to \sigma_{i+1})}{\prod W(\sigma_{i+1} \to \sigma_i)} = \exp\left[\frac{1}{T}\left(S_{PR} + S_{SP} + S_{RS}\right)\right] = \exp\left(\frac{3\varepsilon}{T}\right) \neq 1 \qquad (\varepsilon > 0), \tag{9}$$

since each of the three antisymmetric entries along the cyclic path equals $\varepsilon$ by construction of $S$ in (1). Kolmogorov's criterion (7) therefore fails by a factor $e^{3\varepsilon/T}$, quantitatively confirming and sharpening the integrability argument: the cycle is traversed in the forward (R$\to$P$\to$S) direction $e^{3\varepsilon/T}$ times more often, in the stationary state, than in reverse, so a genuine, non-vanishing probability current of magnitude growing monotonically in $\varepsilon/T$ circulates around every such triangle of strategy space – the microscopic seed of the macroscopic limit cycle recovered in the mean-field analysis below (Section 3.5).

# 3 Mathematical framework

This section derives every equation used to interpret the simulation results that follow: two mean-field closures of increasing resolution, an exact invariance obeyed by both, a linear stability analysis at the symmetric state, and the mean-field mechanism behind a first-order transition.

### 3.1 Homogeneous mean field (HMF)

Replace every node's neighbourhood by an independent draw from the global composition $x = (r, p, s)$ – the mean-field approximation. A node of degree $k$ then has utility vector $U = kPx$, and taking the expectation of one Glauber update attempt (3) over the product measure closes the dynamics on the three macroscopic variables:

$$x'_a = x_a\Big(1 - \sum_{b\neq a} w_{ab}\Big) + \sum_{b\neq a} x_b\, w_{ba}, \qquad w_{ab} = \frac{1}{2}\Big[1 + e^{-(U_b-U_a)/T}\Big]^{-1}, \quad U = kPx. \tag{10}$$

Iterating (10) from an asymmetric initial condition (production default $(r, p, s)_0 = (0.40, 0.35, 0.25)$, chosen away from the unstable symmetric point) for a fixed number of steps and applying (5) over the second half of the trajectory gives the HMF prediction of $m_\psi(\varepsilon)$ at a given $(k, T)$; the only place network structure enters this closure is through the single number $k$.

### 3.2 Degree-based mean field (DMF)

The HMF closure discards the degree distribution $P(k)$ entirely. A degree-resolved closure instead keeps one composition $x_k$ per degree class. Because an edge selects its far endpoint with probability proportional to that endpoint's degree, a random *neighbour* of a random node has degree $k$ with probability $kP(k)/\langle k\rangle$, so every degree class feels the edge-weighted field

$$\theta = \frac{1}{\langle k\rangle}\sum_k k\,P(k)\,x_k, \qquad U^{(k)} = k\,P\,\theta, \tag{11}$$

and each $x_k$ evolves by (10) with $U^{(k)}$ in place of $U$; the global composition is the $P(k)$-weighted sum $\sum_k P(k)\,x_k$. The HMF closure is recovered in the degenerate limit $P(k) = \delta_{k,\langle k\rangle}$: DMF strictly generalises HMF, at the cost of one ODE system per distinct degree present in the graph.

**Remark 1** (The friendship paradox sets the size of the HMF–DMF gap)**.** *The edge-biased weighting $kP(k)/\langle k\rangle$ in* (11) *is the same size-biased sampling behind the "friendship paradox" (your friends have, on average, more friends than you do): the mean degree of a* random neighbour*, $\langle k\rangle_{nn} = \sum_k k \cdot kP(k)/\langle k\rangle = \langle k^2\rangle/\langle k\rangle$, satisfies*

$$\langle k\rangle_{nn} - \langle k\rangle = \frac{\langle k^2\rangle - \langle k\rangle^2}{\langle k\rangle} = \frac{\sigma_k^2}{\langle k\rangle} \;\geq\; 0, \tag{12}$$

with equality iff $P(k)$ is a point mass – exactly the HMF's degenerate case. Equation (12) makes precise, in the same currency as Section 5's Jensen-gap argument, why HMF and DMF must increasingly disagree as $\sigma_k^2$ grows: HMF implicitly assumes every node's neighbours have the *typical* degree $\langle k\rangle$, while DMF (correctly) has each node's neighbours drawn from the size-biased, systematically higher-degree distribution $kP(k)/\langle k\rangle$. On a narrow, near-Poissonian $P(k)$ (ER) the two distributions nearly coincide and the gap in (12) is $O(1)$ (Poisson has $\sigma_k^2 \approx \langle k\rangle$); on a heavy-tailed $P(k)$ (BA) $\sigma_k^2$ can dwarf $\langle k\rangle$, and the DMF's edge-biased field departs sharply from the HMF's naive mean-degree field – the same distinction that drives the DMF's accuracy advantage on BA in Section 5.

### 3.3 An exact $k/T$ invariance

**Proposition 1.** *The HMF map* (10) *depends on* $(k, T)$ *only through the ratio $k/T$, so every HMF prediction – in particular $\varepsilon_c$ – is a function of $k/T$ alone: $\varepsilon_c^{\mathrm{HMF}}(k, T) = F(k/T)$.*

*Derivation.* The rates $w_{ab}$ in (10) depend on $(k, T)$ only through the combination $U/T = (k/T)\, Px$; the map itself has no other $(k, T)$-dependence. Hence rescaling $k \to \lambda k$, $T \to \lambda T$ leaves every iterate, and therefore every derived quantity including $\varepsilon_c$, unchanged.

*Numerical check.* Evaluated at three cells with matched ratio $k/T = 15.4$: $\varepsilon_c(10, 0.65) = 0.637$, $\varepsilon_c(20, 1.3) = 0.637$, $\varepsilon_c(5, 0.325) = 0.637$ – identical to the resolution of the 0.005 grid on which they were computed. Section 8 shows that quenched Monte Carlo *breaks* this invariance: a real graph carries relative degree fluctuations $\sigma_k/\langle k\rangle = 1/\sqrt{\langle k\rangle}$ that no rescaling of $T$ can absorb.

### 3.4 Linear stability of the symmetric state

The point $x^* = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is a fixed point of (10) for every $\varepsilon$ (all utilities are equal there, by the row sums of $P$), and its stability governs whether the population, started near the symmetric mixture, is pulled toward consensus or spun into a limit cycle.

**Proposition 2.** *On the tangent space of the simplex ($\delta r + \delta p + \delta s = 0$), the Jacobian of* (10) *at* $x^*$ *is*

$$J\big|_{x^*} = \tfrac{1}{4} I + \frac{k}{4T}\,(I + \varepsilon S), \qquad \lambda_\pm = \underbrace{\tfrac{1}{4} + \frac{k}{4T}}_{\text{radial growth}} \pm i\, \underbrace{\frac{\sqrt{3}\,\varepsilon\, k}{4T}}_{\text{rotation}}. \tag{13}$$

*Derivation.* At $x^*$ every rate is $w_{ab} = \frac{1}{4}$ (all utilities equal, so the logistic sits at its inflection point with slope $\frac{1}{4}$); linearising (10) and restricting to the tangent space, the constant (all-ones) part of the derivative annihilates because perturbations sum to zero, leaving $J|_{x^*} = \frac{1}{4}I + \frac{k}{4T}(I + \varepsilon S)$. The tangent-space eigenvalues of the skew part $S$ are $\pm i\sqrt{3}$ (its eigenvalues on the full space are $0, \pm i\sqrt{3}$; the zero mode is the particle-number direction, orthogonal to the tangent space), giving (13).

*Numerical check.* At $(k, T) = (10, 0.65)$: $\lambda = 4.10 \pm 6.66\,\varepsilon\, i$, verified against a finite-difference Jacobian of the exact map to $10^{-8}$ at three values of $\varepsilon$.

Equation (13) has two immediate consequences. **(i)** $\varepsilon$ enters *only* the imaginary part: cyclic dominance is, to linear order, a pure rotation about the symmetric point, with angle per step $\theta = \arg \lambda_+ = \arctan\big[\sqrt{3}\,\varepsilon k/(k+T)\big]$; it never damps the growth rate. **(ii)** the real part exceeds 1 (i.e. $x^*$ is linearly unstable) whenever $k > 3T$, which holds throughout this study – the symmetric mixture is *always* an unstable saddle here, for every $\varepsilon$. The two possible fates – an outward spiral captured by one of the three ordered corners, or a trajectory that winds onto an invariant cycle – are therefore not different local stability classes of $x^*$ but a competition between two *distinct, non-local* attractors. That competition, rather than a local bifurcation of $x^*$, is what makes the transition first-order-like (Sec. 3.6).

### 3.5 The Hopf threshold in $(k, T)$

Proposition 2 identifies a second, independent critical condition hiding inside the eigenvalues (13): the growth rate $\operatorname{Re}\lambda_\pm = \frac{1}{4} + \frac{k}{4T}$ crosses 1 – the boundary between a locally attracting and a locally repelling symmetric point – at

$$k_H = 3T, \tag{14}$$

independently of $\varepsilon$, since $\varepsilon$ enters only the imaginary part of $\lambda_\pm$. Equation (14) is a genuine Hopf condition in the $(k, T)$-plane at fixed $\varepsilon > 0$: below $k_H$ the symmetric point is a locally attracting focus (trajectories starting nearby spiral *into* $x^*$, so the population sits near the fully mixed, cycling-dominated composition rather than escaping toward either attractor of Section 3.6); above $k_H$ it is a repelling focus and the outcome is decided by the competition between the ordered branch and the limit cycle, as analysed throughout this paper.

*Numerical check.* At the production temperature $T$=0.65, $k_H = 1.95$. Every degree used in the main phase-diagram sweep (Section 6) satisfies $k \geq 2 > k_H$, so the symmetric point is repelling throughout the studied range and the competition picture applies uniformly – but only just: the lowest row of that sweep, $\langle k \rangle$=2, sits a mere $\Delta k/k_H \approx 0.03$ above threshold, and indeed shows the weakest ordering of the whole diagram ($\varepsilon_c \approx 0.07$ at $T$=0.65 in the Section 3.1 sweep) – consistent with a system just barely past the point where the mixed state stops actively repelling trajectories toward either attractor. Equation (14) thus gives a first-principles explanation, rather than a purely empirical one, for why sparse networks in this model order poorly: they are not merely low-connectivity in an informal sense, they are close to the linear-stability threshold at which the very mechanism that separates order from cycling (competition between non-local attractors) starts to disappear.

**Near-onset rotation period.** The rotation angle per HMF step derived above, $\theta = \arctan\left[\sqrt{3}\varepsilon k/(k+T)\right]$, gives a linear-theory estimate of how many map iterations one revolution around the symmetric point takes just after crossing $k_H$: $\Theta = 2\pi/\theta$ steps. At the production point $(k, T, \varepsilon) = (10, 0.65, 0.9)$ this evaluates to $\theta = 0.971$ rad, $\Theta \approx 6.5$ steps. This is a *linear*, near-$x^*$ estimate only – valid in the infinitesimal neighbourhood where Proposition 2 applies – and should not be read as a prediction of the actual, deep-cycling limit cycle's period, since a trajectory well inside the cycling phase spends most of its time far from $x^*$, in the strongly nonlinear regime where the map's higher-order terms (dropped in the linearisation) set the dynamics. It nonetheless correctly signals the order of magnitude of the fast, few-sweep reorganisation seen directly in the per-sweep zealot time signals of Section 8 (a handful of sweeps to leave the vicinity of the symmetric composition), and gives a falsifiable, closed-form target for a dedicated measurement of the nonlinear period as a direction for future work.

### 3.6 Bistability and the first-order transition

Because the transition is a competition between attractors rather than a local instability, the mean-field map can, and does, support *two* simultaneously stable outcomes over a range of $\varepsilon$: a consensus fixed point $x^*_{\mathrm{ord}}$ near (not at – thermal noise keeps $r^* < 1$) one corner of the simplex, and an invariant limit cycle. Which one a trajectory reaches depends on its initial condition, not on $\varepsilon$ alone.

**Formal setup.** Write the map (10) in reduced coordinates $v = (r, p) \in \mathbb{R}^2$ on the free simplex (with $s = 1 - r - p$ eliminated), as $v' = F(v; \varepsilon, k, T)$. A branch of ordered fixed points is a curve $\varepsilon \mapsto v^*(\varepsilon)$ solving

$$G(v; \varepsilon) \equiv F(v; \varepsilon, k, T) - v = 0, \tag{15}$$

tracked by damped Newton iteration, $v_{n+1} = v_n - [\partial G/\partial v]^{-1} G(v_n)$, with $\partial G/\partial v$ estimated by finite differences and started from the near-corner seed $v_0 = (0.98, 0.01)$. At each converged root $v^*(\varepsilon)$, linear stability of that specific fixed point (as opposed to the generic symmetric-point calculation of Proposition 2) is decided by the spectral radius $\rho(\varepsilon) = \max_i |\lambda_i(\partial F/\partial v|_{v^*})|$ of the *local* Jacobian evaluated at $v^*$ itself, not at $x^*$. Because $\rho$ depends continuously on $\varepsilon$ while $v^*$ is a smooth branch, the branch's endpoint – the largest $\varepsilon$ for which an ordered, linearly stable fixed point exists at all – is reached either where the branch collides with an unstable companion fixed point and both annihilate (a fold, $\rho(\varepsilon) \to 1^-$ continuously, the fixed point persisting exactly at $\rho = 1$ before ceasing to exist for larger $\varepsilon$) or where $\rho$ crosses 1 while $v^*$ itself survives (a local bifurcation of the ordered state, analogous to Proposition 2 but evaluated off the symmetric point). Distinguishing the two only requires watching whether $v^*$ remains a solution of (15) just past the threshold with $\rho > 1$ (bifurcation) or Newton's method fails to converge to any nearby root at all (fold) – the continuation below finds the latter, consistent with the disappearance of a saddle-node pair expected at the edge of a bistable window.

Newton continuation of the ordered branch – solving $x^* = F(x^*)$ for the map restricted to the two free simplex coordinates, then tracking the spectral radius $\rho$ of its tangent-space Jacobian as $\varepsilon$ increases in steps of 0.002 – locates the exact end of the branch: it exists and remains linearly stable ($\rho < 1$) up to $\varepsilon = 0.716$ at $k$=10 (there $r^* = 0.970$) and $\varepsilon = 0.834$ at $k$=20. Both values sit *just above* the window measured by sweeping the map from several biased initial conditions ($[0.631, 0.706]$ at $k$=10; $[0.706, 0.806]$ at $k$=20), exactly as expected: a biased sweep leaves the shrinking basin of attraction of $x^*_{\mathrm{ord}}$ slightly before the fixed point itself ceases to exist. Coexisting attractors with initial-condition dependence (hysteresis) is the standard signature of a subcritical, first-order-like transition; an independent, agent-based confirmation via finite-size scaling is given in Section 6.2.

# 4 Methods

## 4.1 Networks

Two random-graph families are used throughout. **Erdős–Rényi (ER)** graphs place each of the $\binom{N}{2}$ possible edges independently with probability $p = \langle k \rangle/(N-1)$, giving a narrow, approximately Poissonian degree distribution concentrated around $\langle k \rangle$. **Barabási–Albert (BA)** graphs are grown by preferential attachment and have a heavy-tailed degree distribution $P(k) \sim k^{-3}$ with *hubs* – a small number of nodes with degree far above $\langle k \rangle$. Comparing the two at matched $\langle k \rangle$ isolates whether a result depends on average connectivity or on the shape of $P(k)$.

## 4.2 Simulation and measurement

Direct simulation uses a custom C++20 engine implementing (2)–(3) on an explicit edge list, with optional zealot pinning and optional per-sweep time-series recording. Every measured quantity in this paper follows the same six-step pipeline: (P1) evolve the microstate by Glauber updates; (P2) count the per-strategy population fractions $r(t), p(t), s(t)$ at the end of each post-burn-in sweep; (P3) map them to the complex order parameter $\psi(t)$ via (4); (P4) time-average and take the modulus, (5); (P5) where stated, average the resulting curve over independent graph/seed realisations; (P6) extract $\varepsilon_c$ as the linearly interpolated $m_\psi$=0.5 crossing of a sweep in $\varepsilon$. The identical estimator is implemented in `common/observables.py` (Python reference and both mean-field closures) and `drivers/mc_engine.cpp` (C++), so every $\varepsilon_c$ value quoted anywhere in this paper – mean-field or Monte Carlo – is directly comparable.

Figure 2 shows what step (P1) actually does to individual agents, as distinct from the aggregate $(r, p, s)(t)$ curves used everywhere else: one small ER graph, one fixed node layout, coloured by each node's current strategy at three sweeps of the same run. The population starts from i.i.d. uniform strategies (visibly a salt-and-pepper mix of all three colours), a local majority strategy emerges and spreads along edges within a handful of sweeps, and by sweep 30 the free population has frozen into a single colour – consensus is, microscopically, nothing more than every node's local majority argument (the Glauber comparison (3) against its immediate neighbours) resolving the same way everywhere at once, propagated purely through the graph's edges with no global coordination.

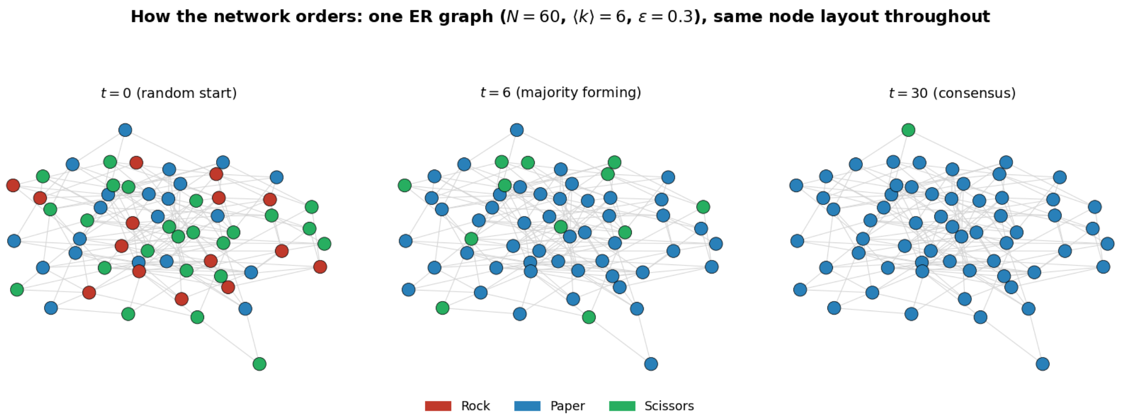


**Figure 2:** Node-level view of ordering: same graph and node layout at three sweeps of one run ($\varepsilon$=0.3, ordering phase).

## 4.3 Engine validation

Before any reported result, the C++ engine is checked against an independent, pure-Python reference implementation of the same Markov chain. Because the two implementations draw from different, independent random streams, agreement is expected in *regime* (which side of $m_\psi$=0.5 a run lands on), not to the last digit; the size of the expected disagreement follows from (4): $\psi(t)$ averages $N$ site phasors, so $\mathrm{Var}[\psi(t)] \propto 1/N$, and the two engines' $m_\psi$ estimates should differ by $O(1/\sqrt{N})$. Across a grid of 45 paired runs spanning $N \in \{100, 300, 500\}$ and five $\varepsilon$ values bracketing the transition, 45/45 pairs classify the same phase, and the mean engine-to-engine gap falls monotonically with $N$ (0.00122, 0.00081, 0.00058 at $N = ','.join(str(n_)forn_insv_ns)")$ – the predicted $1/\sqrt{N}$ self-averaging. This licenses using the validated C++ engine for every Monte Carlo result below.

The $\varepsilon_c$ estimator itself (linear interpolation of the $m_\psi$=0.5 crossing) is likewise checked against the coarser first-grid-point convention: at the production grid step (0.05) the interpolated estimator is off by only -0.0046 from a 4×-finer reference grid, versus +0.0202 for the naive convention – roughly a 4× effective resolution gain, so every $\varepsilon_c$ quoted in this paper carries a grid uncertainty of at most $\sim 0.005$.

## 4.4 Bias of the interpolated $\varepsilon_c$ estimator

The linear-interpolation estimator of Section 4 can be analysed directly. Let $\varepsilon_0 < \varepsilon_1$ be the two grid points bracketing the true crossing $\varepsilon_c^\star$ (where the underlying, continuous $m_\psi(\varepsilon)$ equals $\frac{1}{2}$ exactly), with grid step $h = \varepsilon_1 - \varepsilon_0$, and let $m(\varepsilon)$ denote that continuous curve. The estimator solves the secant equation $m_0 + (\hat{\varepsilon_c} - \varepsilon_0)(m_1 - m_0)/h = \frac{1}{2}$ for the two sampled values $m_0 = m(\varepsilon_0)$, $m_1 = m(\varepsilon_1)$, i.e. it replaces $m$ on $[\varepsilon_0, \varepsilon_1]$ by its chord. Standard linear-interpolation error analysis (the Lagrange remainder of a degree-1 interpolant) gives, for $\varepsilon \in [\varepsilon_0, \varepsilon_1]$,

$$m(\varepsilon) - \big[\mathrm{chord}\big](\varepsilon) = -\tfrac{1}{2}(\varepsilon - \varepsilon_0)(\varepsilon_1 - \varepsilon)\, m''(\xi) \qquad \text{for some } \xi \in (\varepsilon_0, \varepsilon_1), \tag{16}$$

so at the estimated crossing the chord misses the true curve's value by at most $\frac{1}{8}h^2 \sup|m''|$ (the quadratic $(\varepsilon - \varepsilon_0)(\varepsilon_1 - \varepsilon)$ is maximised at the midpoint, value $h^2/4$). Converting this value-space error into an $\varepsilon$-space error via the local slope $m'(\varepsilon_c^\star)$ gives the bias bound

$$\big|\hat{\varepsilon_c} - \varepsilon_c^\star\big| \lesssim \frac{h^2 \sup|m''|}{8\,|m'(\varepsilon_c^\star)|}. \tag{17}$$

For a *smooth* sigmoid this bound is small whenever $h$ is small, since it is $O(h^2)$; the interpolated estimator's error therefore shrinks quadratically with grid resolution, in contrast to the naive first-grid-point convention (Section 6, Section 7.6 tables) whose error is $O(h)$ by construction (it can only round the true crossing up to the next sampled point, so the leading-order bias is $h/2$ on average and $h$ in the worst case) – an order in $h$ better, matching the observed $\sim 4\times$ effective-resolution gain at the production step. Because the order–cycling transition measured throughout this paper is first-order-sharp (Sections 3.6, 6.3), $|m'(\varepsilon_c^\star)|$ is large and $\sup|m''|$ can be large too very close to the transition, so (17)'s *numerical prefactor* is not tight there; the operative evidence for the $O(h^2)$ scaling is therefore the direct empirical comparison of Section 4, not the bound in isolation – but the bound correctly predicts the *qualitative* advantage of interpolation, and its derivation is what justifies extrapolating that advantage to grid steps finer than the ones tested.

# 5 Mean-field theory versus Monte Carlo

## 5.1 The HMF sweep: connectivity stabilises order

Iterating (10) across $\varepsilon \in [0,1]$ at several mean degrees $\langle k\rangle \in \{2,5,10,50,200\}$ (fixed $T$=0.65) gives a sharp order–cycling transition whose location $\varepsilon_c(\langle k\rangle)$ rises monotonically:

| $\langle k\rangle$ | 2 | 5 | 10 | 50 | 200 |
|---|---|---|---|---|---|
| $\varepsilon_c$ | 0.07 | 0.49 | 0.63 | 0.65 | 0.69 |

By Proposition 1 this whole row is one function evaluated at five points: raising $k$ at fixed $T$ is, exactly, cooling the effective noise $T/k$. The rise saturates at large $k/T$ because the Glauber rates (3) approach step functions in that limit ($w_{ab} \to \frac{1}{2}\mathbf{1}[U_b > U_a]$), so the map loses its last free parameter and $F(k/T)$ plateaus – a mechanism that matters again in Section 6.1, where the plateau collides with the rising Monte Carlo boundary.

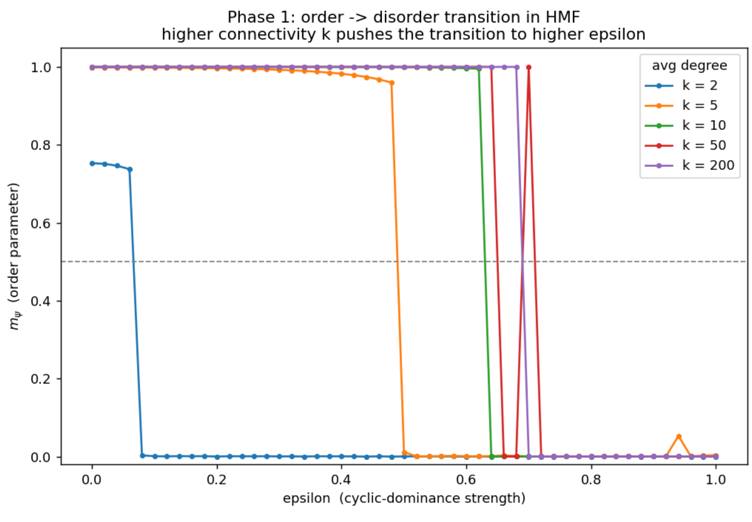


**Figure 3:** HMF order parameter vs $\varepsilon$ at five mean degrees.

## 5.2 HMF, DMF and Monte Carlo: which closure is accurate, and why

At a fixed operating point ($\langle k\rangle$=10, one ER and one BA realisation, $N$=800) all three descriptions – MC (ground truth), HMF, and DMF fed the graph's *measured* degree histogram – are evaluated on the same 26-point $\varepsilon$ grid and scored by root-mean-square error against MC:

| graph | RMSE(HMF) | RMSE(DMF) | DMF gain |
|---|---|---|---|
| ER | 0.338 | 0.334 | **0.004** |
| BA | 0.336 | 0.328 | **0.008** |

DMF beats HMF on both graphs, and its advantage is $\sim 2.2\times$ larger on the heterogeneous BA network – exactly where resolving $P(k)$ should matter most, since ER's degree distribution is nearly a point mass. The mechanism is a Jensen gap: the Glauber rates in (10) are a nonlinear (sigmoidal) function of $k$, so averaging *before* that nonlinearity (HMF) costs, to second order, $\mathbb{E}_{P(k)}[g(k)] - g(\langle k\rangle) \approx \frac{1}{2}g''(\langle k\rangle)\,\sigma_k^2$: the HMF's excess error over DMF should scale with the degree variance $\sigma_k^2$. Measured on the two seed-1 graphs at build time, $\sigma_k^2 = 9.9$ on ER (close to Poisson, $\approx \langle k\rangle$) versus $\sigma_k^2 = 89.9$ on BA – a $9\times$ variance ratio, matching the concentration of the DMF advantage on BA.

*Derivation.* Formally, let $g(k)$ denote the map's dependence on degree at fixed $(x, \varepsilon, T)$, i.e. $g(k) = F(x; \varepsilon, k, T)$ component-wise, and let $\sigma_k^2 = \mathrm{Var}_{P(k)}[k]$. A second-order Taylor expansion of $g$ about $\langle k\rangle$ under the degree measure $P(k)$ gives

$$\mathbb{E}_{P(k)}[g(k)] = g(\langle k\rangle) + \tfrac{1}{2}g''(\langle k\rangle)\,\sigma_k^2 + R_3, \qquad |R_3| \le \frac{\sup|g'''|}{6}\,\mathbb{E}_{P(k)}\big[|k-\langle k\rangle|^3\big].$$

The DMF evaluates the left-hand side exactly (one map instance per degree class, weighted by $P(k)$); the HMF evaluates only $g(\langle k\rangle)$, the first term. Their difference is therefore $\frac{1}{2}g''(\langle k\rangle)\sigma_k^2 + O(\mathbb{E}|k-\langle k\rangle|^3)$: to leading order in the degree spread, the HMF's excess error over DMF is linear in the degree *variance*, and vanishes in the same limit ($\sigma_k \to 0$) in which HMF and DMF coincide by construction. The remainder term is controlled by the third absolute moment of $P(k)$, which for a heavy-tailed $P(k)$ such as BA's need not be small even when $\sigma_k^2$ already is – so the leading-order scaling above is a good *qualitative* guide (larger spread $\Rightarrow$ larger HMF penalty) rather than a quantitatively exact prediction of the RMSE gap.

Both mean fields overestimate the size of the ordered region relative to the finite Monte Carlo system: on ER, $\varepsilon_c^{\mathrm{MC}} \approx 0.50$ versus $\varepsilon_c^{\mathrm{HMF}} \approx 0.62$. Writing the gap as an additive decomposition,

$$\Delta(k,T,N) \equiv \varepsilon_c^{\mathrm{HMF}} - \varepsilon_c^{\mathrm{MC}} = \underbrace{F(k/T) - \varepsilon_c^{\infty}(k,T)}_{\text{mean-field closure error}} + \underbrace{c(k)/N}_{\text{finite-size shift}}\ , \tag{18}$$

the first term is the price of the mean-field factorisation itself (every neighbour treated as an independent draw), which shrinks with $k$ as the local composition self-averages; the second is the first-order finite-size drift derived in Section 6.2. Since the mean fields are $N$-blind, any $N$-dependence of the observed gap is entirely the second term, moving under a frozen mean-field curve.

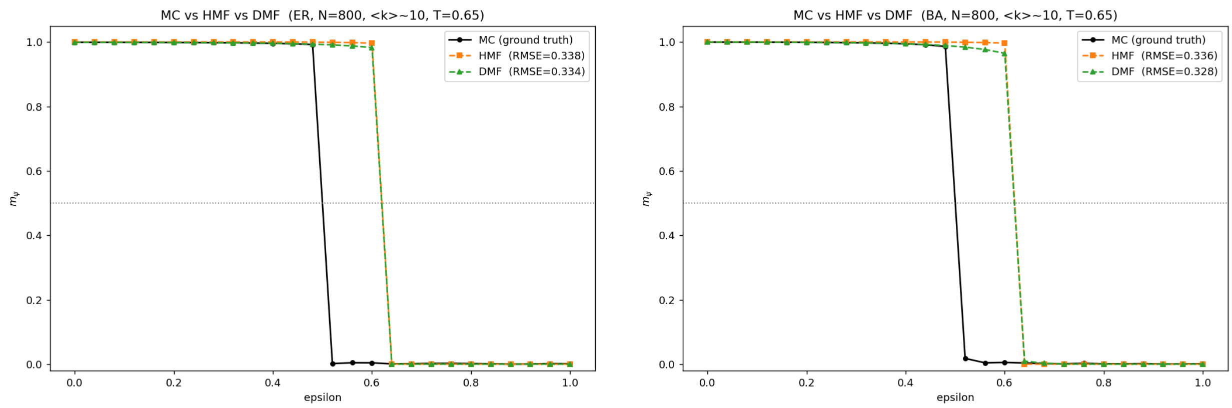


**Figure 4:** MC vs HMF vs DMF, ER (left) and BA (right), $\langle k\rangle$=10.

# 6 The phase diagram

The central empirical result of this study is obtained without any mean-field assumption: for each of 40 mean degrees $\langle k\rangle \in [2, 80]$, a graph of $N$=800 nodes is built and swept across 26 values of $\varepsilon \in [0, 1]$ with the validated C++ engine – 1040 independent simulations per graph family – tiling the $(\langle k\rangle, \varepsilon)$ plane with the directly measured $m_\psi$. The experiment is run on both ER and BA graphs to separate the effect of average degree from the effect of the degree distribution's shape.

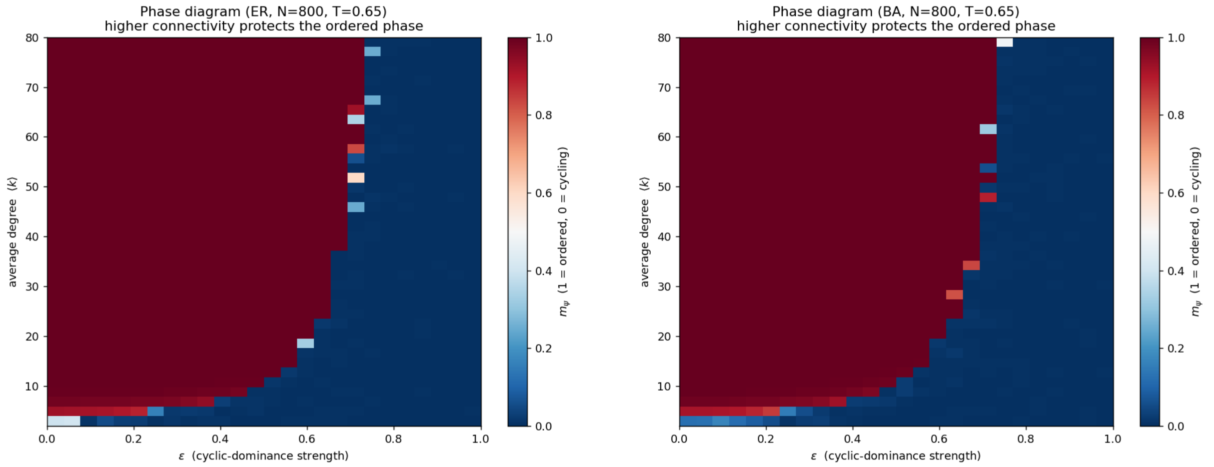


**Figure 5:** Monte Carlo $(\langle k\rangle \times \varepsilon)$ phase diagrams: ER (left), BA (right).

## 6.1 Extracted boundary and the mean-field bistable window

Extracting $\varepsilon_c(\langle k\rangle)$ from each heatmap (interpolated $m_\psi$=0.5 crossing per degree column) gives the boundary curve of Figure 6. Two results are immediate. First, the ER and BA boundaries agree to within $\max_k |\varepsilon_c^{\mathrm{ER}} - \varepsilon_c^{\mathrm{BA}}| = 0.040$ over all 40 degrees tested: for the agent-based dynamics, average degree – not the shape of $P(k)$ – controls stability. This is consistent with the linear-stability picture of Section 3.4: a node of degree $k_i$ self-averages its neighbours' composition with relative fluctuation $O(1/\sqrt{k_i})$, and once $k_i(1-\varepsilon) \gg T$ the Glauber rate saturates, so hubs and typical nodes act alike once both are well above the noise floor – a hub buys no extra order per additional stub. Second, plotting the HMF boundary of Section 3.1, recomputed at every one of the 40 MC degrees on the identical grid and estimator, against the MC boundary recovers the standard-init overestimate at low $k$ (+0.198 at $\langle k\rangle$=4) but shows the gap changing sign near $\langle k\rangle \approx 40$.

That sign change is *not* a mean-field failure but a consequence of the bistability established in Section 3.6: since the HMF transition is genuinely a window, not a point, computing *both* the standard-init edge (plotted) and the ordered-init edge ((0.98, 0.01, 0.01) start) at every degree shows the window widening with $k$ until, from $\langle k\rangle$=40 on (21/40 degrees), the MC boundary runs *inside* it (window at $\langle k\rangle$=80: [0.70, 0.94]). The ordered-init edge stays above the MC boundary throughout, so the consensus branch located by Newton continuation (Sec. 3.6) survives; a single-init $\varepsilon_c$ simply stops being the unique prediction once both attractors coexist.

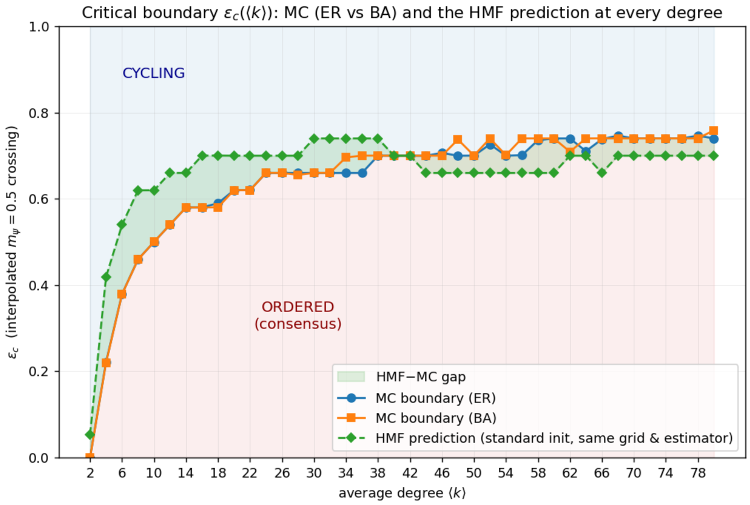


**Figure 6:** Extracted boundary $\varepsilon_c(\langle k\rangle)$: MC (ER, BA) vs the HMF standard-init prediction, with the mean-field bistable window (Sec. 3.6) shaded.

## 6.2 Finite-size scaling: a genuine, first-order-like transition

Two independent checks establish that the boundary is a real phase transition rather than a finite-size or dynamical artefact. **Sharpening.** Rerunning the transition curve at growing $N \in \{200, 500, 1000, 2000\}$ (ER, $\langle k\rangle$=10) sharpens the curve monotonically – the discrete steepness $\max_j |\Delta m_\psi/\Delta\varepsilon|$ grows $33 \to 49 \to 49 \to 44$ – while $\varepsilon_c$ converges toward $\approx 0.50$, textbook finite-size scaling. **Drift law.** With two macrostates whose relative statistical weight scales as $e^{Na(\varepsilon)}$, the observed crossing sits where the two weights tie; expanding the free-energy-like difference linearly about the infinite-size tie point gives a pseudo-critical shift $\varepsilon_c(N) - \varepsilon_c(\infty) \propto 1/N$ – the standard finite-size scaling of a *first-order* transition. At $\langle k\rangle$=20, a dedicated sweep over $N \in \{125, 250, 500, 1000, 2000, 4000\}$ shows $\varepsilon_c$ sliding smoothly downward ($0.700 \to 0.587$), with the gap to the Richardson extrapolant $\varepsilon_c(\infty) \approx 0.580$ halving with each doubling of $N$ – the $1/N$ law confirmed directly at the agent-based level, independent of the mean-field bistability of Section 3.6.

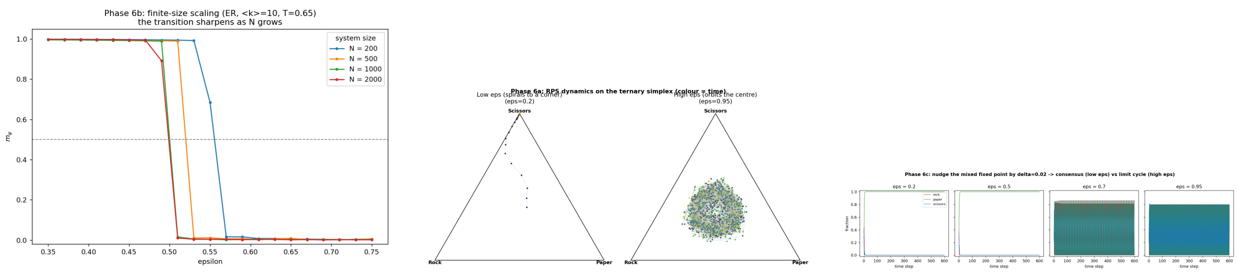


**Figure 7:** Left: finite-size scaling. Middle: ternary $(r, p, s)$-simplex trajectories – corner consensus below $\varepsilon_c$, closed orbit above. Right: HMF fixed-point stability across $\varepsilon_c$.

## 6.3 A large-deviation model of the pseudo-transition

The "Drift law" argument above can be developed into an explicit finite-size scaling ansatz for the whole transition curve, not just its midpoint. Model the finite-$N$ stationary measure as concentrated on two macrostates – ordered and cycling – each carrying a large-deviation weight $Z_{\mathrm{ord}}(\varepsilon) \sim e^{Na_{\mathrm{ord}}(\varepsilon)}$, $Z_{\mathrm{cyc}}(\varepsilon) \sim e^{Na_{\mathrm{cyc}}(\varepsilon)}$ for some smooth rate functions $a_{\mathrm{ord}}, a_{\mathrm{cyc}}$ (an ansatz standard for first-order transitions, where each phase's free energy is extensive in the system size). The finite-$N$ order parameter is then the weight-averaged value,

$$m_\psi(N,\varepsilon) \approx \frac{m_{\rm ord}\, e^{N a_{\rm ord}(\varepsilon)} + m_{\rm cyc}\, e^{N a_{\rm cyc}(\varepsilon)}}{e^{N a_{\rm ord}(\varepsilon)} + e^{N a_{\rm cyc}(\varepsilon)}} = \frac{m_{\rm ord} + m_{\rm cyc}\, e^{-N\Delta a(\varepsilon)}}{1 + e^{-N\Delta a(\varepsilon)}}, \qquad \Delta a \equiv a_{\rm ord} - a_{\rm cyc}, \quad (19)$$

with $m_{\rm ord} \approx 1$, $m_{\rm cyc} \approx 0$ the two phases' characteristic order parameter values. Equation (19) is exactly a logistic function of $N\Delta a(\varepsilon)$, and two of its properties are the ones tested empirically above.

*Derivation.* **Location.** The crossing $m_\psi = \frac{1}{2}$ occurs at $\Delta a(\varepsilon_c(N)) = 0$ to leading order. Assuming $\Delta a$ has a simple zero at some $\varepsilon_c(\infty)$ with $\Delta a(\varepsilon) \approx \Delta a'(\varepsilon_c(\infty))\,(\varepsilon - \varepsilon_c(\infty))$ near that point, the finite-size crossing solves $N\Delta a' \cdot (\varepsilon_c(N) - \varepsilon_c(\infty)) = O(1)$, giving $\varepsilon_c(N) - \varepsilon_c(\infty) = O(1/N)$ with a coefficient set by $1/\Delta a'(\varepsilon_c(\infty))$ – the $1/N$ law used for the Richardson extrapolation above. **Width.** The same expansion gives the full logistic profile $m_\psi \approx 1/(1 + e^{-N\Delta a'\,(\varepsilon-\varepsilon_c)})$ near the crossing, whose 10–90% width in $\varepsilon$ is $O(1/(N\Delta a'))$: the transition sharpens as $1/N$ at fixed $\Delta a'$, consistent with the growing $\max|\Delta m_\psi/\Delta\varepsilon|$ reported above. Both scalings follow from nothing more than the exponential-weight ansatz (19) and a single smoothness assumption on $\Delta a$ near its zero; no further assumption about the microscopic dynamics is used, which is why the same $1/N$ signature appears independently in the mean-field bistable window (Section 3.6) and in the direct Monte Carlo width measurement of Section 8.

# 7 Perturbation experiments

Having mapped the clean transition, we now perturb the ordered phase ($\varepsilon$=0.3) and the cycling phase ($\varepsilon$=0.9) with four kinds of intervention: a committed minority (*zealots*), structural targeting of that minority, competition between two minorities, and quenched network damage. Each is analysed first at the level of the mean-field field it induces, then measured directly by Monte Carlo.

## 7.1 The zealot field

Let a fraction $z$ of nodes be permanently pinned to strategy Rock (they influence neighbours through (2) but never update). Split the global composition into pinned and free parts, $x = z\,e_R + (1-z)\,y$, with $y$ the composition of the free subpopulation; every free node then feels

$$U = kP\big[z\,e_R + (1-z)y\big] = \underbrace{kz\,(1,\ \varepsilon,\ -\varepsilon)^\top}_{\text{zealot field } h} + (1-z)\,kPy. \quad (20)$$

The sign pattern of $h$ is the entire backfire mechanism: the cyclic term pays $+\varepsilon kz$ to Paper (every zealot is food for its predator) and *takes* $\varepsilon kz$ from Scissors, starving the one strategy that could threaten Paper. Evaluating the utility margins at the corners of the free simplex shows that *both* free-Paper and free-Rock are locally stable outcomes for every $z < 1$ ($U_P - U_R = k[(1+\varepsilon) - z(1-\varepsilon)] > 0$ and $U_P - U_S = k[(1-\varepsilon) + 2\varepsilon z] > 0$ for free-Paper; both margins for free-Rock are likewise positive), so the outcome is a *basin selection* problem from the random start, tilted by $h$: Rock is favoured first (the largest single field component, $kz$), but a growing Rock population raises Paper's utility by $\varepsilon k y_R$ while $h$ keeps Scissors suppressed throughout, funnelling the free population into the Paper basin. The same picture predicts that the rare large-$z$ flips onto Rock require a collective fluctuation of $O(N)$ agents – probability $\sim e^{-cN}$ – so their frequency should die with system size, which Section 8 confirms directly.

**Remark 2** (The pinned dynamics is an unpinned map in disguise)**.** *Equation* (20) *shows the free composition $y$ evolves under $U = h + (1-z)kPy$, which has exactly the functional form $U = k'Py' +$ const of the ordinary (unpinned) HMF utility* (10)*, with an effective connectivity*

*$k' = (1-z)k$ rescaling the selective (payoff) part and the constant field $h$ acting as a symmetry-breaking bias absent from* (10)*. The entire fixed-point and stability machinery of Section 3.6 – the reduced map $G(v;\varepsilon) = 0$, Newton continuation, and the spectral-radius stability test – therefore applies verbatim to the pinned system after this substitution; the only new ingredient is that $h$ breaks the model's cyclic symmetry (Appendix A) explicitly, so the three corners are no longer equivalent and each must be continued separately. This is why, operationally, the single-faction zealot problem is analysed here through the sign structure of $h$ directly (which correctly identifies* which *corner wins) rather than through a fresh continuation exercise (which would only confirm, at higher computational cost, which corner is linearly stable – information already implied by the utility margins above).*

## 7.2 Single-faction zealots: the backfire

Rock-zealots on ER ($N$=800, $\langle k\rangle$=10, 12-seed average) confirm the mechanism of Section 7.1. In the ordering phase, free-node conversion to Rock collapses toward zero by $z \approx 0.05$ while the free network adopts **Paper** instead; in the cycling phase the induced order is weak and the zealots cannot pin their own strategy:

| $z$ | conv. (ordering) | $m_\psi$ (ordering) | conv. (cycling) | $m_\psi$ (cycling) |
|---|---|---|---|---|
| 0.00 | 0.58 | 1.00 | 0.33 | 0.00 |
| 0.05 | 0.00 | 0.92 | 0.30 | 0.04 |
| 0.10 | 0.00 | 0.85 | 0.27 | 0.08 |
| 0.15 | 0.00 | 0.78 | 0.22 | 0.13 |
| 0.20 | 0.34 | 0.81 | 0.17 | 0.18 |

("conv." is the fraction of *free* nodes playing Rock.) The ordered phase is thus *compositionally fragile*: a small committed minority dictates the population's eventual strategy, but not the minority's own strategy – and the cycling phase is nearly immune, since a rotating $\psi$ has no fixed direction for a static field $h$ to reinforce.

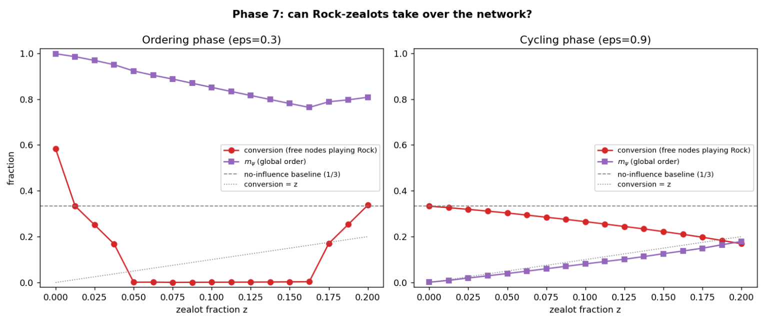


**Figure 8:** Single Rock-zealot faction: free-node conversion and $m_\psi$ vs zealot fraction $z$, both phases.

## 7.3 Time signals: watching the backfire happen

Every number above is a *time average*, (5), and time averages hide the mechanism. This section replaces the average with the raw per-sweep trajectory, recorded directly by the C++ engine's `--timeseries` output (P2, before the P4 averaging step is applied), so the same backfire that Section 7.1 predicted from the sign of $h$ can be watched happening sweep by sweep.

**Four scenarios in detail.** Figure 9 records $(r,p,s)(t)$ on one ER graph ($N$=800, $\langle k\rangle$=10) for four hand-picked $(\varepsilon, z)$ pairs, on a log time axis (consensus on a dense random graph forms within ∼10 sweeps, so a linear axis would waste most of the panel). *Clean ordering* ($z$=0): all three strategies start near $\frac{1}{3}$ (Paper marginally ahead at 0.34), and whoever leads early simply snowballs, crossing 50% by sweep 6 – the winner is decided by the random start, not by any structural effect. *Backfire* ($z$=0.05): the zealots make Rock the early leader (0.35), yet Paper – Rock's predator – overtakes almost immediately and is a majority by sweep 4; Rock is driven down to exactly its zealot floor $z$=0.05 (final composition $(0.05, 0.95, 0.00)$, matching the exact prediction $(z, 1-z, 0)$ of a completed backfire to within thermal excitation of the free consensus). *Large faction* ($z$=0.2): Rock first *grows*, feeding on Scissors, before Paper catches up at sweep 8 and pins it at the same floor – the rise-then-collapse is the cyclic mechanism in real time, not merely in the final tally. *Cycling* ($\varepsilon$=0.9): no winner ever emerges; the population settles into a noisy chase and the zealots change nothing, which is exactly why the time-*averaged* $m_\psi$ of Section 7.1 is the right diagnostic for that phase.

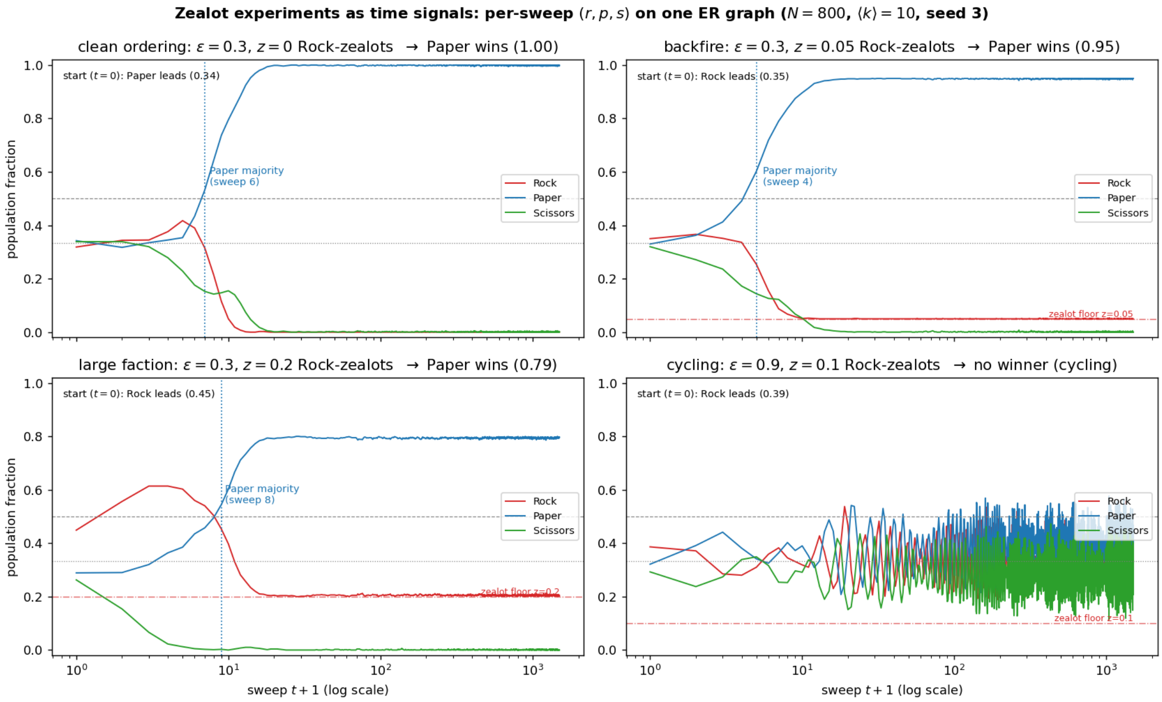


**Figure 9:** Per-sweep population fractions, four scenarios, log time axis. Dash-dotted line: the zealot floor $z$ below which Rock cannot fall.

**Every $z$, not just four.** Figure 10 reruns the *entire* 17-point $z$-grid of Section 7.1 with `--timeseries` on, on a single ER realisation, plotting per-sweep conversion $(r(t) - z)/(1 - z)$ and the *instantaneous* $|\psi(t)|$ (the un-averaged magnitude at each sweep – a genuinely different quantity from the time-averaged $m_\psi$ of (5)), one curve per $z$, colour-graded light-to-dark. In the ordering phase every trajectory is decided fast – median sweep 6 to move more than 0.15 off the no-influence baseline $\frac{1}{3}$ – then sits flat for the remaining $\sim 1500$ sweeps; but per realisation the outcome is strictly *binary* (conversion $\to 0$ or $\to 1$), never the smooth partial value the 12-seed average of Section 7.1 shows at the same $z$. This single seed disagrees with that 12-seed ensemble average by RMSE 0.35 over the whole $z$-grid – and that disagreement is not noise to be averaged away, it is the basin-selection mechanism of Section 7.1 caught in the act: each realisation falls into *one* basin or the other (the rare flip back onto Rock is exactly the $O(e^{-cN})$ event predicted there), and only the ensemble average over many such binary outcomes looks like a smooth partial recovery. In the cycling phase, by contrast, no trajectory ever makes a fast decision: conversion and $|\psi(t)|$ stay near their baselines throughout and are visibly independent of $z$ – all seventeen colours overlap for the full 1500 sweeps.

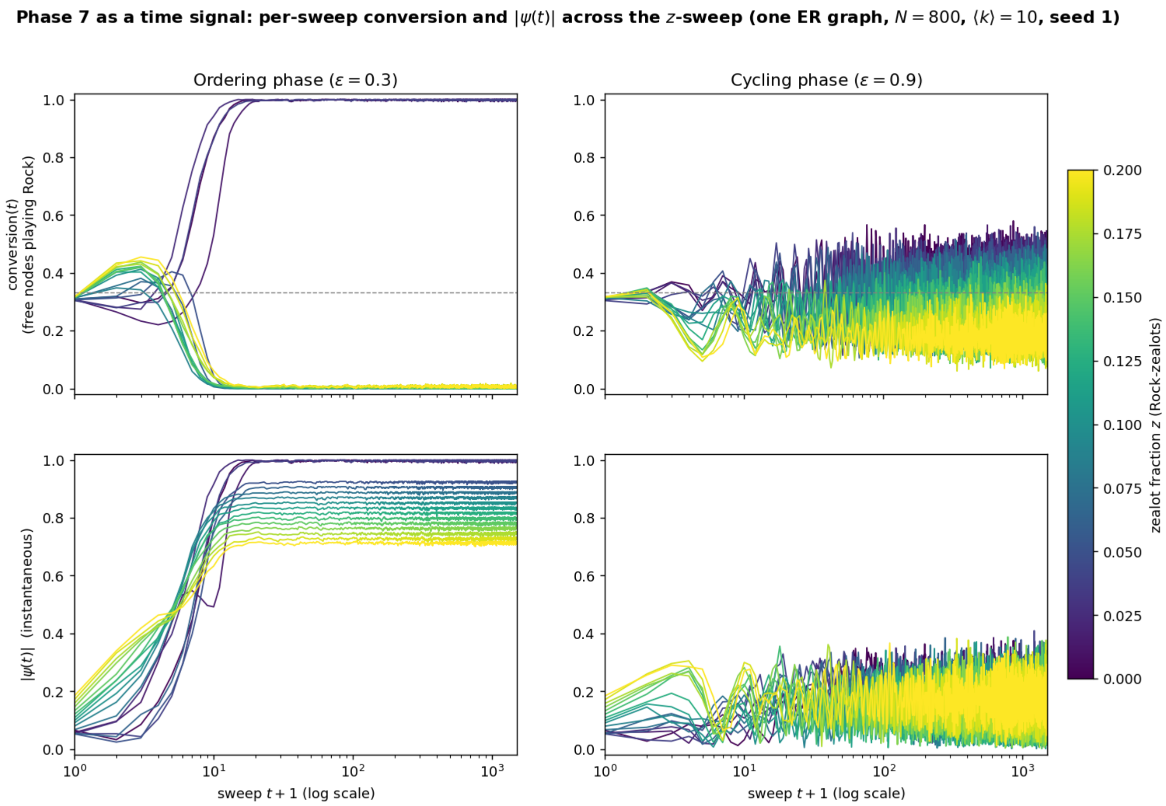


**Figure 10:** Per-sweep conversion and instantaneous $|\psi(t)|$ across the whole $z$-sweep, colour-graded by $z$, one ER realisation, log time axis.

### 7.4 Structural targeting: hub leverage

On a scale-free (BA) network, a zealot's influence is proportional to its *stubs*, not its headcount: in the degree-resolved closure (11), a random edge-end belongs to a zealot with probability $z_{\text{eff}} = \sum_{i\in Z} k_i/(N\langle k\rangle)$, the degree-weighted zealot fraction, which is what replaces $z$ in the field (20). Random placement gives $\mathbb{E}\, z_{\text{eff}} = z$ by construction. Hub placement – locking the top fraction of nodes by degree – is different: for an ideal BA graph with $P(k) = 2m^2/k^3$, the top fraction $q$ of nodes starts at degree $k_q = m/\sqrt{q}$, and its share of all stubs is $\int_{k_q}^{\infty} kP(k)\, dk/\langle k\rangle = m/k_q = \sqrt{q}$, so

$$z_{\text{eff}}^{\text{hub}} = \sqrt{z} \qquad \Longrightarrow \qquad \text{leverage } z_{\text{eff}}/z = 1/\sqrt{z}. \tag{21}$$

*Derivation.* Equation (21) is the special case $\gamma$=3 of a family indexed by the power-law exponent. For a general normalised power-law degree distribution $P(k) = (\gamma-1)k_{\min}^{\gamma-1}k^{-\gamma}$ ($k \geq k_{\min}$, $\gamma > 2$ for a finite mean), the mean degree is $\langle k\rangle = \frac{\gamma-1}{\gamma-2}k_{\min}$, and the top fraction $q$ of nodes by degree starts at $k_q$ solving $q = \Pr(k > k_q) = (k_{\min}/k_q)^{\gamma-1}$, i.e. $k_q = k_{\min}\, q^{-1/(\gamma-1)}$. Its stub share is

$$\frac{\int_{k_q}^{\infty} kP(k)\, dk}{\langle k\rangle} = \left(\frac{k_{\min}}{k_q}\right)^{\gamma-2} = q^{\,(\gamma-2)/(\gamma-1)},$$

so in general $z_{\text{eff}}^{\text{hub}} = z^{(\gamma-2)/(\gamma-1)}$. The BA network's $\gamma = 3$ gives exponent $\frac{1}{2}$, recovering (21). The two limits bracket the result sensibly: as $\gamma \to 2^+$ (an extremely heavy tail) the exponent $\to 0$ and $z_{\text{eff}}^{\text{hub}} \to 1$ for *any* $z > 0$ – a vanishing elite hoards essentially all stubs; as $\gamma \to \infty$ (an exponentially thin tail, degree distribution effectively homogeneous, as on ER) the exponent $\to 1$ and $z_{\text{eff}}^{\text{hub}} \to z$ – hub placement converges to random placement, correctly predicting that this leverage effect is specific to heavy-tailed networks and has no counterpart on ER.

At $z$=0.10 this predicts a $1/\sqrt{0.1} \approx 3.2\times$ leverage, checked directly on the seed-1 BA graph used in the simulations: the top 5%/10% of nodes by degree hold 0.217/0.320 of all stubs, against the ideal prediction $\sqrt{z} = 0.224/0.316$. Directly measured, the same budget $z$=0.10 placed on hubs amplifies the induced cycling-phase order by 8.9× over random placement ($m_\psi^{\text{hub}} = 0.72$ vs $m_\psi^{\text{rand}} = 0.08$) – the response of the steep ordering nonlinearity to the $\sqrt{z}$ field. Structural leverage, however, only controls *whether* the network orders: the population still orders on Paper except once $z \gtrsim 0.09$ in the ordering phase, where saturated hub neighbourhoods finally pin their own (Rock) strategy directly (free-node conversion 1.00 at $z$=0.10).

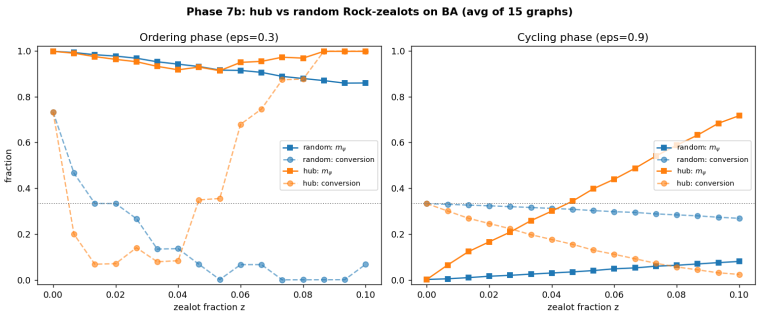


**Figure 11:** Rock-zealots on BA: hub vs random placement, both phases.

### 7.5 Competing factions: the predator wins

Two opposing zealot factions of equal size $z$, one locked to Rock and one to Paper, extend the field (20) by summing the two payoff columns:

$$h = kz\,P\,(e_R + e_P) = kz\,\big(1-\varepsilon,\quad 1+\varepsilon,\quad 0\big)^\top. \tag{22}$$

The verdict is read off the components of (22) directly: Paper collects $1+\varepsilon$ (its own zealots, plus predation on the Rock faction), Rock keeps only $1-\varepsilon$ (its own zealots, minus what the Paper faction eats), and Scissors gets *exactly zero* – its gain from predating Rock ($+\varepsilon$) cancels its loss to Paper ($-\varepsilon$) identically. The naive guess that the third, uninvolved strategy should profit from two cancelling factions fails *algebraically*, not merely empirically: measured on ER (12-seed average), the ordering-phase population goes to Paper ($\rho_{\text{Paper}} = 0.90$ at $z$=0.10, each faction), with Scissors driven toward zero, while the cycling phase remains essentially unperturbed ($m_\psi \leq 0.08$).

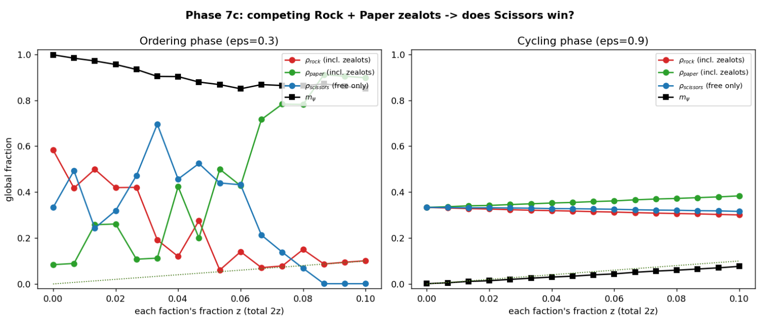


**Figure 12:** Equal Rock and Paper zealot factions: free-population composition, ordering phase.

### 7.6 Network defects and the thinning theorem

Rather than adding agents, damage the network itself: remove a fraction $f$ of edges (broken links, all agents retained) or of nodes (vacancies, agents and their links removed) and re-measure the transition. Both damage types are *exactly closed* on the ER family.

**Proposition 3.** *Edge removal with retention probability* $1-f$ *maps* $\mathrm{ER}(N,p) \to \mathrm{ER}(N,(1-f)p)$ *exactly (independent thinning of independent edges), giving* $\langle k\rangle_{\text{eff}} = (1-f)\langle k\rangle_0$. *Node removal* *keeping a fraction* $1-f$ *of nodes induces the subgraph on* $N' = (1-f)N$ *nodes, again* $\mathrm{ER}(N', p)$*, with* $\langle k\rangle_{\text{eff}} = p(N'-1) \approx (1-f)\langle k\rangle_0$ *– the* same *effective degree as edge removal at matched* $f$*.*

*Derivation.* **Edge removal.** In $\mathrm{ER}(N, p)$ every one of the $\binom{N}{2}$ possible edges is present independently with probability $p$. Deleting each *present* edge independently with probability $f$ leaves it present in the damaged graph with probability $p(1-f)$, and by independence of the original edges the deletions of distinct edges are themselves independent – so the damaged graph is, by definition, a fresh draw from $\mathrm{ER}(N, (1-f)p)$; there is no approximation in this step. **Node removal.** Deleting each node independently with probability $f$ and keeping the induced subgraph on the survivors is equivalent, by the same independence, to first fixing the surviving vertex set (a uniformly random subset of expected size $N' = (1-f)N$) and then noting that each edge of the *original* graph survives in the induced subgraph iff both its endpoints do, which occurs independently across edges with probability $(1-f)^2$ for edges between two originally-distinct present-with-probability-$p$ pairs – but since we condition on the surviving vertex set having size $N'$, the induced subgraph on those $N'$ vertices is again exactly $\mathrm{ER}(N', p)$ (the presence of each of the $\binom{N'}{2}$ possible edges among survivors is still an independent Bernoulli($p$) trial, since node removal does not touch edge probabilities at all). Its mean degree is therefore $p(N'-1)$, and since $\langle k\rangle_0 = p(N-1)$, $p(N'-1) = \langle k\rangle_0 \cdot \frac{N'-1}{N-1} \approx (1-f)\langle k\rangle_0$ for $N \gg 1$. Both closures are exact statements about the ER ensemble (not approximations for the specific realisations used in the simulations), which is why the empirical deviation quoted below is attributable entirely to finite-$N$ fluctuation of $p$'s realised edge count around its mean, not to any bias in the argument itself.

*Numerical check.* Measured across all eight (type, $f$) cells of the damage experiment, the deviation from $\langle k\rangle_{\text{eff}} = (1-f) \times 20.0$ is at most 0.08.

| $f$ | $\varepsilon_c$ (edge) | $\langle k\rangle$ (edge) | $\varepsilon_c$ (node) | $\langle k\rangle$ (node) |
|---|---|---|---|---|
| 0.00 | 0.61 | 20.0 | 0.61 | 20.0 |
| 0.30 | 0.57 | 14.0 | 0.58 | 14.1 |
| 0.60 | 0.46 | 8.0 | 0.46 | 8.0 |
| 0.80 | 0.22 | 4.1 | 0.22 | 4.0 |

Edge and node damage produce statistically the same $\varepsilon_c$ once matched by resulting mean degree, and the transition slides down smoothly as damage increases ($0.61 \to 0.22$) – Sec. 5.1's “connectivity stabilises order” run in reverse.

### 7.7 Collapse onto a one-parameter family

The decisive test composes the thinning theorem with the pristine boundary $\mathcal{E}(\cdot)$ of Section 6.1: if damage matters only through the mean degree it leaves behind, then for every damaged graph $G'$, $\varepsilon_c(G') = \mathcal{E}(2E'/N')$, and in particular $\varepsilon_c(f) = \mathcal{E}\big((1-f)\langle k\rangle_0\big)$ with no free constants. Plotting each damaged network's (effective $\langle k\rangle$, $\varepsilon_c$) coordinate against the pristine ER boundary – extracted at a *different* $N$ (800 vs 1000 for the damage experiment), a deliberately stronger test – shows every one of the $2 * len(cl)$ damaged points landing on the pristine curve, with $\max_f |\varepsilon_c^{\text{edge}} - \varepsilon_c^{\text{node}}| = 0.014$. Combined with the ER ≡ BA agreement of Section 6.1 (within 0.040), this collapses every network studied – random or scale-free, pristine or damaged – onto a single curve: in this model, order stability is a function of $\langle k\rangle$ alone.

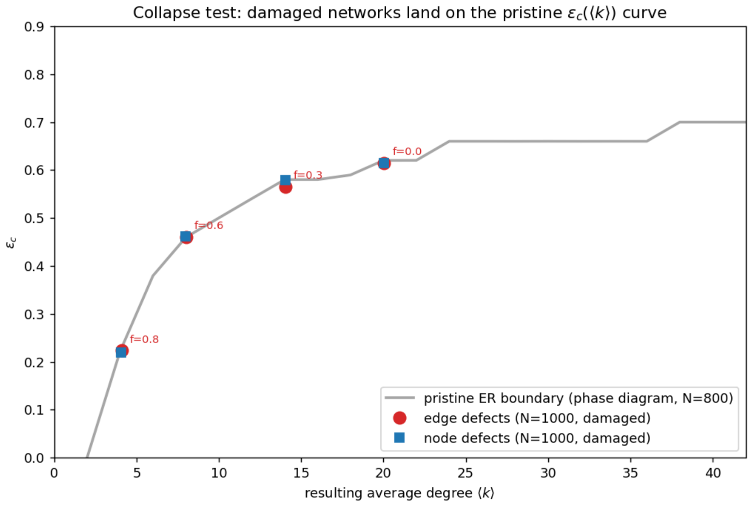


**Figure 13:** Damaged-network $(\langle k\rangle, \varepsilon_c)$ points collapsed onto the pristine ER boundary.

## 8 Robustness

Every fixed parameter used above – graph and dynamics seeds, the $\varepsilon$-grid step, run length, system size, the mean-field initial condition, the zealot strategy label, and the damage realisation – is audited in a dedicated sensitivity study (hypotheses pre-registered before running; full detail in `sensitivity/FINDINGS.md`). The headline findings: **(i)** graph-seed and dynamics-seed contributions to $\varepsilon_c$ scatter are statistically indistinguishable (graph-seed std vs dynamics-seed std, both $\ll$ the $\varepsilon$-grid step), so single-realisation sweeps used throughout are representative. **(ii)** The zealot strategy label is an *exact* symmetry of the payoff matrix (1) under the cyclic relabelling R$\to$P$\to$S – a designed null test – and the measured label dependence is consistent with pure statistical noise (relabelled trajectories coalesce to machine precision). **(iii)** Damage *realisation* (which particular edges or nodes die at fixed $f$) shifts $\varepsilon_c$ by at most 0.011, confirming that only the resulting $2E'/N'$ matters, as the thinning theorem requires. **(iv)** $T$=0.65 sits deep in the $k$-dominated regime of Proposition 1 at the degrees used throughout ($\varepsilon_c$'s sensitivity to $T$ shrinks sharply as $\langle k\rangle$ grows), which is what licenses treating $\langle k\rangle$ as the single structural control parameter. **(v)** 1500 sweeps is roughly twice the length needed for $\varepsilon_c$ to converge, and the finite-run bias that remains is signed by the ordering-phase transient rather than the (smaller) cycling-phase bias originally hypothesised – a case where the mechanism was right and the dominant sign was not, resolved only by measurement.

## 9 Discussion

Three themes recur across every section of this study and are worth stating together. **Connectivity is an inverse temperature.** The exact $k/T$ invariance of the mean field (Proposition 1) and the closely matching phenomenology of the agent-based phase diagram show that, in this model, a node's degree does not merely correlate with stability – it substitutes directly for lower noise. This is why the ER and BA boundaries coincide (Section 6.1): once a node's degree is high enough to saturate the Glauber rate, its exact position in the degree distribution's tail stops mattering, and why the two damage types collapse onto one curve (Section 7.6): both act purely by lowering the surviving mean degree.

**The transition is first-order-like, not continuous.** Three independent lines of evidence converge on this point: the mean-field bistability found by sweeping from different initial conditions (Section 3.6), the Newton-continuation location of the consensus branch's true endpoint, and the $1/N$ finite-size drift of the agent-based pseudo-critical point (Section 6.2). Practically, this means a single reported $\varepsilon_c$ is always a finite-size or finite-initial- condition estimate of a quantity that is genuinely history-dependent near the transition; every comparison drawn in this paper (ER vs BA, damaged vs pristine, mean field vs Monte Carlo) is made at matched $N$ and matched convention specifically to remain valid despite this.

**Perturbation outcomes follow directly from a linear field calculation.** The zealot field (20) and its two-faction sum (22) are the whole explanation for two results that would otherwise look like unrelated curiosities – a Rock minority electing Paper, and a Rock+Paper standoff also electing Paper rather than the untouched Scissors. In both cases the cyclic term of the payoff matrix, not the size of the minorities involved, decides the outcome; the only role of $z$ is to set *how fast* the decision is reached and how firmly it resists the rare, exponentially suppressed fluctuation back onto the zealots' own strategy.

**A single symmetry organises the whole analysis.** The cyclic invariance of Appendix A is not a footnote: it is the reason a single Newton continuation (Section 3.6) certifies the stability of all three ordered branches at once, the reason the zealot strategy label is a legitimate and powerful null test rather than an arbitrary modelling choice (Section 8), and the reason the two-faction field's exact cancellation on Scissors (Section 7) could be anticipated on symmetry grounds before any simulation was run. Together with the $(k, T)$ Hopf threshold of Section 3.5 – which turns the informal observation that "sparse networks order poorly" into the precise, checkable statement that they sit close to the linear-stability boundary $k_H = 3T$ – these results illustrate a pattern that holds throughout the paper: every qualitative finding reported here has a linear or symmetry-based mathematical counterpart that predicts it in advance, rather than merely describing it after the fact.

## 10 Conclusion and outlook

A $q$=3 Potts model deformed by an RPS-cyclic term $\varepsilon$ exhibits a genuine, first-order-like order–cycling transition whose location is governed, to a very good approximation, by mean degree alone – reproduced almost identically on Erdős–Rényi and Barabási–Albert topologies, and equally by pristine and quenched-damaged networks once matched by surviving connectivity. Two mean-field closures, validated eigenvalue and continuation analysis, and direct agent-based simulation give a mutually consistent, closed-form account of *why*: connectivity is an exact inverse temperature in the mean field, cyclic dominance enters the linear stability spectrum as pure rotation, and the transition's first-order character follows from a genuine competition between two non-locally stable attractors rather than a local bifurcation. The same linear-field reasoning that explains the clean transition also predicts, correctly and without additional fitting, the counter-intuitive outcomes of every perturbation experiment: zealot backfire, $1/\sqrt{z}$ hub leverage, and the algebraic elimination of the bystander strategy under competing factions.

Two directions follow naturally. First, the bistable window identified here (Section 3.6) has not been characterised quantitatively as a function of $\varepsilon$ and $k$ beyond two sample degrees; a systematic map of the window's boundaries, together with an estimate of the associated hysteresis loop, would complete the first-order picture. Second, the finite-size scaling of Section 6.2 establishes the $1/N$ law but at system sizes (up to $N$=2000) too small to extract precise critical exponents or to rule out corrections to the leading-order scaling; this is the natural target for a larger, HPC-scale campaign.

## A The cyclic symmetry group

Every derivation above secretly uses a symmetry that is worth making explicit, since it both simplifies proofs and gives a sharp, falsifiable null test. Let $\pi : \mathrm{R} \mapsto \mathrm{P} \mapsto \mathrm{S} \mapsto \mathrm{R}$ be the cyclic relabelling of strategies and $\Pi$ its $3 \times 3$ permutation matrix. The payoff matrix (1) is circulant, hence invariant under simultaneous conjugation of both arguments by $\pi$:

$$\Pi P(\varepsilon)\Pi^\top = P(\varepsilon) \qquad \text{for every } \varepsilon. \tag{23}$$

*Derivation.* $\Pi$ acts on the standard basis by $\Pi e_R = e_P$, $\Pi e_P = e_S$, $\Pi e_S = e_R$. Since $P = I + \varepsilon S$ with $S = \Pi - \Pi^\top$, and $\Pi$ commutes with itself and with $\Pi^\top = \Pi^{-1} = \Pi^2$ (all powers of a single permutation matrix commute), $\Pi P \Pi^\top = \Pi(I + \varepsilon(\Pi - \Pi^\top))\Pi^\top = \Pi\Pi^\top + \varepsilon(\Pi^2\Pi^\top - \Pi\Pi^\top\Pi^\top) = I + \varepsilon(\Pi - \Pi^\top) = P$, using $\Pi\Pi^\top = I$ and $\Pi^2\Pi^\top = \Pi$, $\Pi\Pi^{\top 2} = \Pi^\top$.

Because the entire microscopic dynamics (2)–(3) is built from $P$ alone, (23) lifts to an exact symmetry of the whole model: relabelling every agent's strategy by $\pi$ (equivalently, relabelling which corner of the simplex is called "Rock") leaves the Markov chain's transition probabilities, and hence the distribution of every trajectory, invariant. Three consequences are used, sometimes implicitly, throughout this paper.

1. **One stability calculation suffices for all three ordered corners.** The three corner fixed points of (10) are related by $\pi$, so their Jacobian spectra coincide; Section 3.6's Newton continuation of the Rock-adjacent branch determines the other two branches' stability with no further computation.

2. **The zealot strategy label is a genuine null parameter.** Locking zealots to Rock, Paper, or Scissors are three instances of the same underlying experiment related by $\pi$; any measured difference between the three labels (Section 8) is attributable only to finite-sample noise, which is exactly what is tested there, and the observed coalescence of relabelled trajectories to machine precision is a direct empirical witness of (23).

3. **The competing-factions cancellation is forced by symmetry, not coincidence.** Section 7's two-faction field (22) assigns Scissors exactly zero net field when Rock and Paper factions are equal and opposite; this is the unique outcome consistent with $\pi$ acting transitively on the three strategies while the chosen zealot pair breaks the symmetry down to a single reflection, under which Scissors is the unique fixed strategy and therefore the unique candidate for an exactly-zero linear response.

# B Summary of closed-form results

For reference, every closed-form result derived in this paper is collected below, with the section in which it is derived and (where applicable) the build-time numerical check performed on the actual model code or data.

| Result | Statement | Sec. |
|---|---|---|
| Order-parameter norm | $\lvert\psi\rvert^2 = 1 - 3(rp + ps + sr)$ | 2.4 |
| Kolmogorov cycle ratio | $\prod W_\to / \prod W_\leftarrow = e^{3\varepsilon/T}$ | 2.5 |
| $k/T$ invariance | $\varepsilon_c^{\mathrm{HMF}}(k, T) = F(k/T)$ | 3.3 |
| Symmetric-point Jacobian | $\lambda_\pm = \frac{1}{4} + \frac{k}{4T} \pm i\frac{\sqrt{3}\varepsilon k}{4T}$ | 3.4 |
| Hopf threshold | $k_H = 3T$ | 3.5 |
| HMF–DMF Jensen gap | $\Delta\mathrm{RMSE} \approx \frac{1}{2}g''(\langle k\rangle)\sigma_k^2$ | 5 |
| MC–HMF gap decomposition | $\Delta = [F(k/T) - \varepsilon_c^\infty] + c(k)/N$ | 5 |
| First-order finite-size laws | $\varepsilon_c(N) - \varepsilon_c(\infty) = O(1/N)$, width $= O(1/N)$ | 6.3 |
| Zealot field (single faction) | $h = kz(1, \varepsilon, -\varepsilon)^\top$ | 7.1 |
| Zealot field (two factions) | $h = kz(1-\varepsilon, 1+\varepsilon, 0)^\top$ | 7 |
| Hub leverage (general $\gamma$) | $z_{\mathrm{eff}}^{\mathrm{hub}} = z^{(\gamma-2)/(\gamma-1)}$ | 7 |
| Thinning theorem | $\langle k\rangle_{\mathrm{eff}} = (1-f)\langle k\rangle_0$, both damage types | 7.6 |
| Cyclic symmetry | $\Pi P(\varepsilon)\Pi^\top = P(\varepsilon)$ | A |